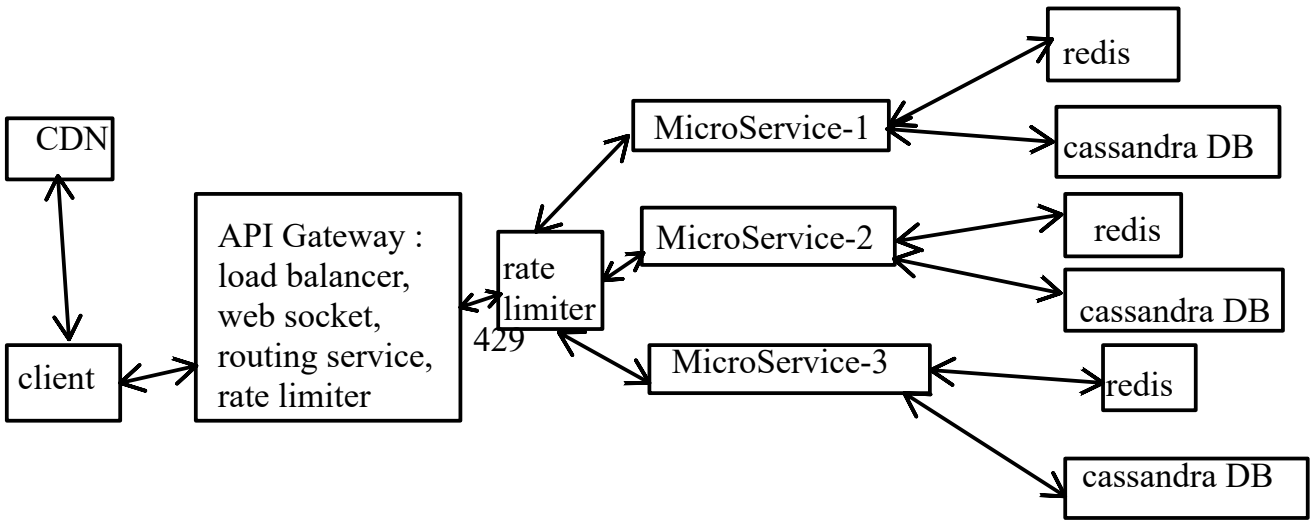
Rate Limiter :

Functional requirements:
rate limit should be able to limit incoming traffic for a particular threshold ,
all the requests crossing the threshold should be discarded

Non-functional requirements:
data-partitioning,
load blanaicng,
fault tolerance,
vertical scalability,
horizontal scalabillity,
low latency,
high throughput,
CAP theorem (Here eventual consistency is fine , but here we need 99.99% i.e Availability>>Consistency)

Core Entities:
user,
request,
response,

API :
POST:request/url
{
userID,
{requestURL}
}
GET:/search/{URL}

HLD: